

INTRODUCTION

The HUSB311 evaluation board EVB_HUSB311_BLA has provide the necessary passive components needed by the HUSB311. It is designed to work with different USB Type-C Port Manager (TCPM). The TCPM can control the HUSB311 via the SDA, SCL, and INT_N pins. After a couple configurations in the TCPM, the HUSB311 can function as a source, sink or dual-role port. All pins has test points and connects to the P1 connector in the schematic

DESCRIPTION

The HUSB311 is a USB Type-C port controller that complies with the latest USB Type-C and PD standards. The HUSB311 integrates a complete Type-C Transceiver including the Rp and Rd resistors. It does the USB Type-C detection including attach and orientation. The HUSB311 integrates the physical layer of the USB BMC power delivery protocol to allow up to 100W of power delivery.

FEATURES

- Dual-Role PD Compatible
- Attach/Detach Detection as Host, Device or DRP
- Current Capability Definition and Detection
- Cable Recognition
- Dead Battery Support
- Ultra-low Power Mode for Attach Detection
- Simple I²C Interface with AP or EC
- BIST Mode Supported
- Two I²C addresses
- 9-Ball WL-CSP and 14-Lead QFN Packages
- E-fuse IP

POWER SUPPLY

The power supply on VDD pin is 2.8V ~ 5.5V. A 0.1uF capacitor is recommended on VDD pin. The capacitor is C5 in the schematic. The external power supply for VDD should connect to the J3 connector.

I²C COMMUNICATION

The TCPM can communicate with the HUSB311 through the I²C by connecting the I²C master signal to the SDA and SCL pins. The I²C address of the HUSB311 can be set as 0x4E or 0x3E. The 0x4E is the default address value.

The SDA and SCL pins can be pull up to VDD pin by R1, R2 separately if the I²C master has no pull up resistors.

INT_N

The INT_N is open drain type interrupt output used to prompt the TCPM to handle the events reported by the HUSB311. The INT_N can be pull up to VDD by R5 if the TCPM has no pull up resistor.

CC PINS

The CC1 and CC2 pins are connected to the Type-C receptacle J1 on the board. The HUSB311 demo contains the cReceiver capacitance specified in the USB PD specification for the CC pins which is 330pF. These capacitors are C6 and C7 in the schematic.

VBUS

The VBUS is connected to the Type-C receptacle J1. One 0.1uF capacitor C11 is on VBUS. The C11 can be removed in typical application.

VCONN

It is a regulated input pin to be switched to correct CC pin as VCONN to power the Type-C full-featured cables and other accessories. The power supply on VCONN is 3.0V ~ 5.5V. One 0.1uF capacitor is recommended on the VCONN pin. The capacitor is C10 in the schematic. The external power supply for VCONN should connect to the J2 connector.